

SAPG STOCK ANALYSIS

ARIMA & VAR forecasting models for stock price prediction

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AGENDA

01 CONTEXT

02 DATA & PREPROCESSING

03 MODEL EXPLANATION

04 ORANGE IMPLEMENTATION

05 MODEL VERDICT

06 REFLECTION

07 CONCLUSION

01

CONTEXT

01 CONTEXT

SAP SE
(EQUITIES: SAPG)

OPEN

PRICE
(CLOSING)

1/1/2019 - 12/31/2025

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02

DATA & PREPROCESSING

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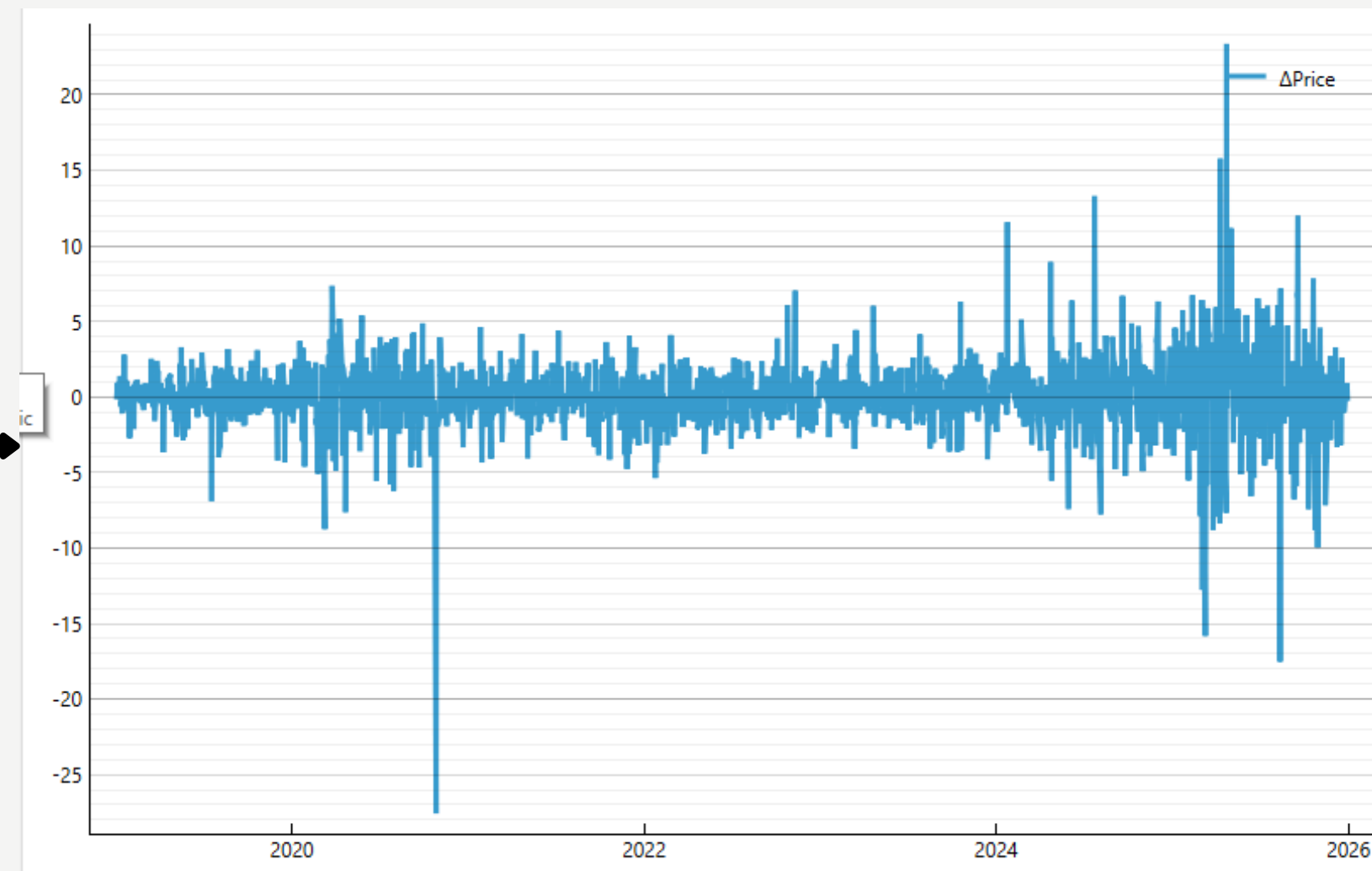
02 DATA & PREPROCESSING



Data Profile:

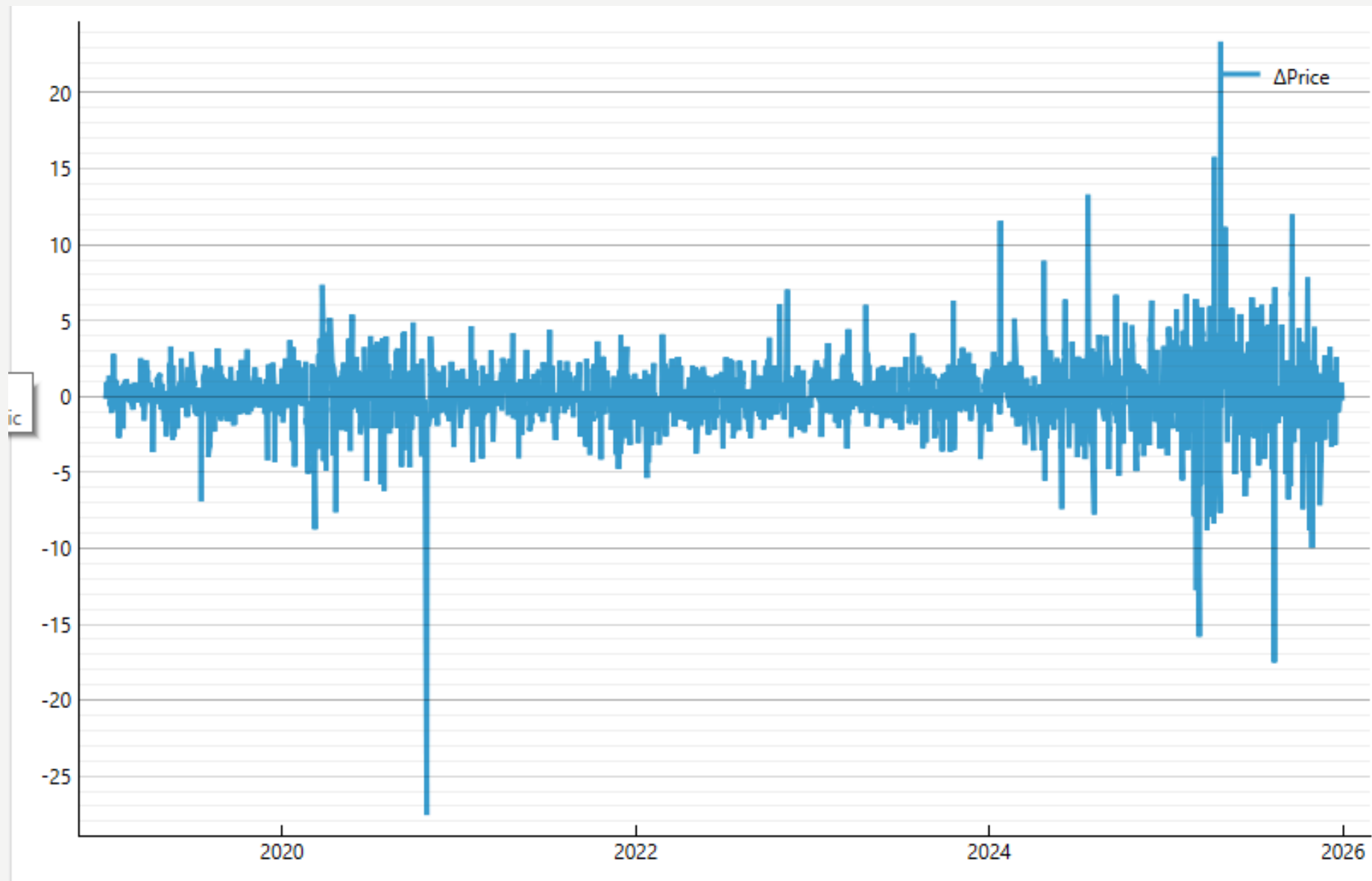
- *SAP Stock market prices*
- *Date Range: 01.01.2019 - 31.12.2025 (6 years)*
- *1779 data points (time frame: daily)*
- *Missing Values where stock-market was not open*

First order difference ($d = 1$)

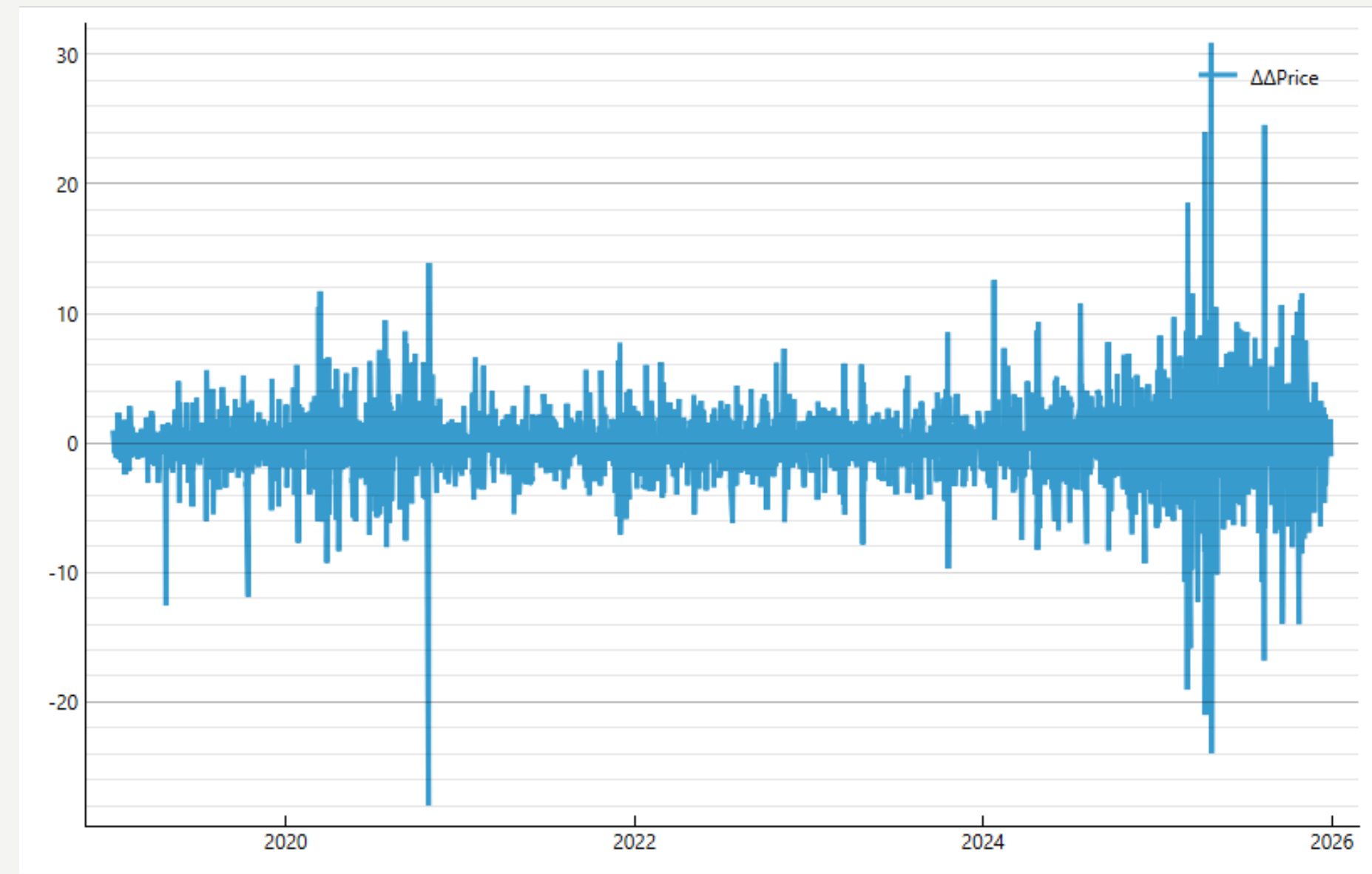


$$\Delta y = y - (y-1)$$

$$d = 1$$



$$d = 2$$



03

MODEL EXPLANATION

03 MODEL EXPLANATION

VAR

- *takes multiple variables into account, by forming a vector for each variable*
- *each vector (variable) consists of its value, lagged value, other variables lagged values and an error value*

ARIMA

AR : Autoregression means that values are evaluated based on previous values

I : Integrated just means that we need to remove stationary

MA : Moving Average means that values are evaluated based on previous error

(1,1,0)

Comparison

VAR

Good:

- *Multiple related variables*
- *no assumptions on type of data*

Bad:

- *needs large amounts of data to make accurate predictions*
- *assumes linear data*

ARIMA

Good:

- *Single variable*
- *reliable with various kinds of data*

Bad:

- *unreliable for seasonal data, assumes stationary data*
- *assumes linear data*

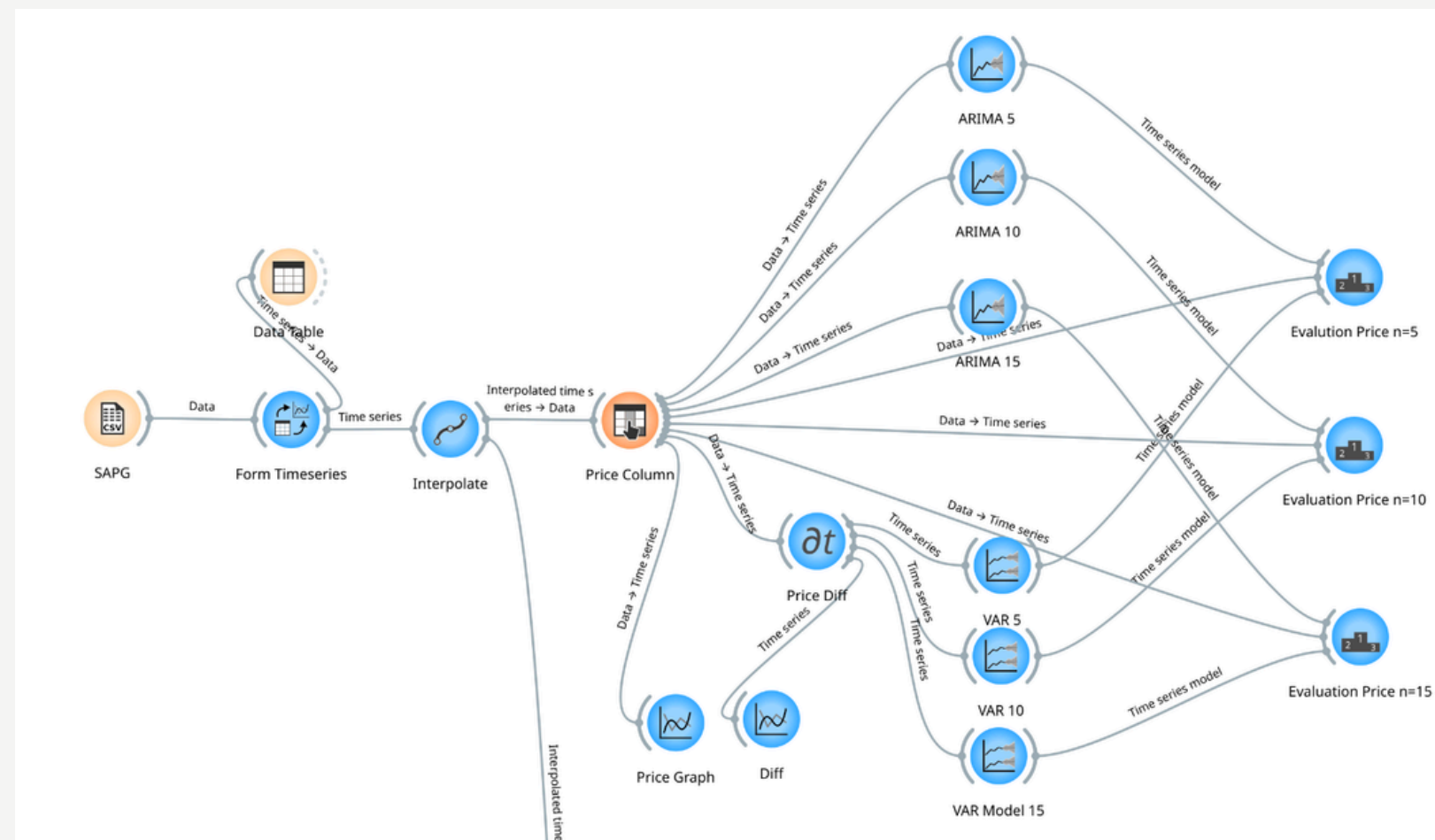
04

ORANGE IMPLEMENTATION

04 ORANGE IMPLEMENTATION

WORKFLOW & FORECASTS

- Orange workflow and forecasts at 5, 10, and 15 days for both ARIMA and VAR then evaluate the model on each time horizon pair



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RMSE & MAE COMPARISON

- PRICE (Close/Adj Close)

Time horizon (n)	RMSE (ARIMA / VAR)	MAE (ARIMA / VAR)
5	6.15 / 6.29	2.95 / 2.98
10	8.96 / 9.06	4.71 / 4.44
15	12.3 / 12.7	7.02 / 6.58

- OPEN

Time horizon (n)	RMSE (ARIMA / VAR)	MAE (ARIMA / VAR)
5	5.26 / 5.51	2.83 / 1.85
10	8.46 / 8.50	4.53 / 3.90
15	11.7 / 12.2	6.51 / 5.83

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MODEL VERDICT

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05 MODEL VERDICT

	Close Price (ARIMA wins)		Open Price (VAR wins)	
Time horizon (n)	RMSE (% Lower than VAR)	MAE (% Lower than VAR)	RMSE (% Lower than ARIMA)	MAE (% Lower than ARIMA)
5	2.20%	1.00%	4.50%	34.60%
10	1.20%	-6.10%	0.50%	13.90%
15	3.90%	-6.70%	-4.1.%	10.40%

※ % Lower Error Formula: $(\text{ARIMA} - \text{VAR}) / \text{ARIMA} \times 100$

Inferences :

- On average across all horizons, ARIMA perform better on predict closing price with 2.4% lower RMSE, while VAR preform better on predicting opening price with 19.6% lower MAE.*

REFLECTION

- We only used two variables.
- Both models assume the relationship between prices stays the same over time.
- Testing across multiple time windows would give more reliable results.

CONCLUSION

Closing Price: ARIMA consistently wins on RMSE
(6.15 vs 6.29 at 5 days, 8.96 vs 9.07 at 10 days, 12.3 vs 12.8 at 15 days).

Opening Price: ARIMA wins RMSE
(5.26 vs 5.51 at 5 days, 8.46 vs 8.50 at 10 days, 11.7 vs 12.2 at 15 days)

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